PROJECT VISION

# Pancreatic cancer image analysis

A research foundation for a future multimodal clinical platform

## Connect morphology, tumor biology and clinical context.

H&E; pathology images + matched EMT assays + independently confirmed stage

TODAY

### Runnable research prototype

LONG-TERM AIM

### Evidence-based support for multidisciplinary review

**Research only. No real patient cohort or clinically validated model is included.**

# Three different questions, three evidence needs

Scope: pathologist-confirmed pancreatic ductal adenocarcinoma (PDAC).

01

## Cancer stage

How far has disease spread?

Full staging requires clinical, radiological and pathological information. A primary-tumor tile cannot establish whole-body extent.

02

## EMT phenotype

What epithelial/mesenchymal features are present?

Use matched, tumor-aware molecular or spatial assays. A cross-sectional prediction is not a measure of progression.

03

## Therapeutic context

What should the care team review?

Resectability, diagnosis, fitness, organ function, molecular findings and patient goals guide discussion.

Sources: NCI PDQ [1]; EMT consensus [2]. Full references on slide 20.

# Begin with an auditable research workbench

A narrower first product makes the scientific question testable.

01

## Primary user

Pathology and translational research teams studying image associations with a predefined EMT assay.

The first use is retrospective feasibility analysis.

02

## Core hypothesis

Patient-level morphology may contain reproducible information associated with EMT labels and broad stage labels.

This remains an unproven hypothesis.

03

## Success definition

Reproducible external performance beyond simple clinical/site baselines, interpretable failures and acceptable workflow burden.

Clinical benefit needs a later study.

# A working baseline with explicit boundaries

Completed software versus proposed clinical-platform capabilities.

| Domain | Available now | Next development |
| --- | --- | --- |
| Images | Single-frame H&E; tile QC and RGB summary features. | Whole-slide ingestion, annotations and validated tumor-region selection. |
| Models | Patient-level stage classifier and EMT regressor. | Compare pathology encoders, multiple-instance learning and multimodal models. |
| Interface | Local tile upload, JSON report and context panel. | Hospital integration, role-based review and controlled report sign-off. |
| Evidence | Synthetic workflow, four tests and dashboard smoke check. | Real cohort, external validation and prospective clinical studies. |

# Build a cohort before building a clinical claim

Labels must be independent of model outputs and aligned to the same specimen/timepoint.

01

## Images + provenance

Reviewed H&E; tumor tiles; patient, specimen and site identifiers.

Record stain, scanner, physical resolution, specimen type and pretreatment status.

02

## Clinical reference

Confirmed PDAC diagnosis; cTNM/pTNM basis and staging edition in the cohort database.

Keep treatment timing, procedure and resectability distinct.

03

## Biological reference

Matched EMT assay with prespecified normalization and tumor-cell localization.

Measure assay reliability and stromal contamination. Use one harmonized score definition.

**Current manifest: patient_id | image_path | stage | emt_score | emt_assay | site_id**

# From reviewed tiles to a patient-level research report

One specimen/timepoint per patient in the current implementation.

01
**Reviewed tiles**

>

02
**QC + 18 features**

>

03
**Patient averaging**

>

04
**Two random forests**

## Split by patient before fitting: 60% training / 20% validation / 20% test

Outputs: stage-associated votes + EMT assay association + feature-range warning + JSON report.
The model does not diagnose cancer, determine clinical stage or select therapy.

# Software pipeline verified on artificial data

SYNTHETIC ONLY - no clinical or biological performance claim.

01

## 80 fictional patients

160 artificial texture tiles.

48 train / 16 validation / 16 test patients.

No real pancreatic tissue images are included.

02

## Held-out demo metrics

Stage balanced accuracy: 100%

EMT MAE: 0.0128 on the artificial 0-1 scale.

The generator deliberately encodes labels in color.

03

## What this proves

Training, artifact saving, patient splitting, evaluation and inference can run end to end.

It does not show ability to stage cancer or measure EMT.

**Four automated tests passed; dashboard rendering and clinical-context interaction also passed.**

# Make every output traceable and qualified

Illustrative report from an in-training synthetic sample - not independent validation.

| Report field | Current demo value | Required interpretation |
|---|---|---|
| Stage association | Class I; vote fraction 0.9917. | Uncalibrated forest vote; clinical stage remains undetermined. |
| EMT association | 0.7635; assay: SYNTHETIC_DEMO. | Artificial cross-sectional score; not a percentage of cells undergoing EMT. |
| Progression | Reported as not determined. | Requires longitudinal matched samples and assay validation. |
| Clinical context | Selected independently by clinician-confirmed resectability. | Educational discussion only; no image-driven prescription. |

# Validate EMT in malignant cells, not just tissue color

A future image model needs reliable molecular ground truth.

01

## Define the endpoint

Prespecify the marker/signature panel, score mapping and assay repeatability.

EMT is heterogeneous and context dependent; a single marker is insufficient.

02

## Localize the signal

Distinguish malignant epithelial cells from fibroblasts and other stromal cells.

Use spatial, multiplex or single-cell evidence where feasible.

03

## Study change separately

Collect paired baseline, on-treatment and progression specimens.

Control batch and sampling effects; test longitudinal models before labeling change as progression.

Source: Yang et al., EMT research consensus [2].

# Support an informed multidisciplinary discussion

Educational pathway categories; actual care remains patient-specific.

| Clinician-entered status | Discussion domain | Clinical review |
|---|---|---|
| Resectable | Surgical assessment and perioperative systemic therapy discussion. | Review stage, fitness, anatomy, pathology and patient preferences. |
| Borderline resectable | Neoadjuvant therapy and reassessment discussion. | Multidisciplinary review determines the pathway. |
| Locally advanced | Systemic therapy and selected local approaches. | Discuss response, symptoms and appropriate trials. |
| Metastatic | Systemic and eligible biomarker-directed options; symptom care. | Molecular assessment, goals and trial eligibility inform review. |

Source: NCI professional pancreatic cancer treatment summary [1].

# Proposed architecture for a clinical-grade service

Future architecture - these production services are not implemented today.

| | |
|---|---|
| **Clinical systems** | Pathology slides + annotations \| radiology/PACS \| EHR \| molecular laboratory |
| **Governed data layer** | Deidentification \| identity linkage \| consent/access rules \| provenance \| versioned labels |
| **Analysis services** | Tile QC \| image encoder/MIL \| optional CT model \| assay alignment \| calibrated model outputs |
| **Review workspace** | Evidence view \| uncertainty/abstention \| clinician review \| signed report \| audit trail |
| **Operations** | Model registry \| monitored deployment \| drift review \| incident response \| controlled updates |

Integration candidates: DICOM/DICOMweb and HL7 FHIR, subject to hospital capabilities and local requirements.

# Design around accountable human review

The research model supplies evidence; the clinical team owns interpretation.

01

## Before analysis

Confirm diagnosis and intended use.

Verify patient/specimen match, timing, image suitability and assay provenance.

Reject unusable or unsupported inputs.

02

## At review

Display evidence provenance, uncertainty, quality failures and model version.

Separate image associations from confirmed clinical findings. Require reviewer sign-off.

03

## After review

Retain an auditable report and record corrections.

Monitor failures, workload and disagreement.

New model versions require controlled evaluation and release.

# Move forward through evidence gates

Illustrative 36+ month plan; timing depends on cohort access, results and intended use.

| Planning horizon | Work package | Deliverable and decision gate |
|---|---|---|
| 0-6 months | Research foundation | Protocol, ethics/data agreements, assay definition, pilot cohort and QC. Gate: usable labels and repeatable assays. |
| 6-12 months | Retrospective development | Compare baseline/encoder models and confounder controls. Gate: reproducible patient-level benefit. |
| 12-24 months | External + silent study | Lock model; evaluate unseen centers and prospective workflow. Gate: prespecified performance and usable review. |
| 24-36+ months | Clinical translation | Impact study, deployment controls and applicable review. Gate: evidence and authorizations for the defined use. |

# Measure reliability before measuring adoption

Specify numerical acceptance criteria with clinicians and statisticians before the study.

01

## Analytical performance

Independent patient/site/time splits; no test-driven normalization or tuning.

Classwise error, calibration, EMT error and patient-bootstrap uncertainty. Test against assay repeatability.

02

## Generalizability

Held-out institutions, scanners and specimen types.

Check shortcuts, subgroup uncertainty, missing labels, rejected cases and distribution shift. Compare clinical-only baselines.

03

## Clinical usefulness

Prospective silent evaluation first, then a suitable impact study.

Assess review time, errors, clinician interpretation and patient-relevant benefit. Stage agreement alone is insufficient.

# Treat lifecycle management as part of the product

Exact obligations depend on jurisdiction, intended use and device classification.

01

## Governance + security

Define purpose, access and retention with each partner.

Role-based access, encryption, audit trails, security testing and incident ownership.

Confirm local privacy obligations.

02

## Quality + safety

Maintain requirements, risk analysis, traceability and verified releases.

Specify abstention behavior, human factors, version history and rollback.

Research tooling is not clinical clearance.

03

## Regulatory planning

Obtain qualified advice early for each intended market.

Confirm study evidence and submission requirements; assess model-change controls.

No approval or timeline is assumed.

Reference: FDA AI-enabled device lifecycle draft guidance, January 2025 [3]; draft, not a binding approval pathway.

# A clinical partnership, supported by engineering

Staffing and resource categories are planning proposals, not committed funding.

01

## Core partnership

Clinical lead: pancreatic oncology/surgery.

Pathology lead and molecular/EMT scientist.

Biostatistician, data steward and patient/public input.

02

## Build + operate

ML engineer, data engineer and platform engineer.

Annotation/QC capacity; product and clinical-integration support.

Quality, regulatory and security expertise as scope grows.

03

## Sustainable funding

Early: research grants and institutional partnerships.

Later: carefully scoped service contracts or licensed validated modules.

Budget data/assays, staff, compute, studies and maintenance.

# Define stop conditions before expanding scope

A failed hypothesis is a useful outcome when the study can detect it.

| Risk | Failure mode | Mitigation / decision |
| --- | --- | --- |
| Stain/site shortcuts | Model learns scanner or sampling procedure instead of biology. | Use external centers and confounder baselines; stop the claim if performance collapses. |
| Weak EMT labels | Bulk stromal signals or noisy assays dominate the target. | Use malignant-cell evidence and repeatability checks; revise the endpoint if unreliable. |
| Misinterpretation | Votes are mistaken for diagnosis, progression or treatment advice. | Review labels, uncertainty and human factors; block unsupported clinical outputs. |
| Operational failure | Identity mismatch, drift or uncontrolled model updates. | Audit linkage, monitor failures, version deployments and provide rollback. |

# Turn the concept into a credible study

Immediate deliverables for a hospital or research collaboration.

01

## Days 1-30

Agree one intended research use and endpoint.

Appoint clinical/pathology leads; map data access and ethics requirements.

Freeze cohort and assay dictionaries.

02

## Days 31-60

Assemble a deidentified pilot and review tumor tiles.

Audit specimen timing, molecular pairing and duplicate identities.

Run baseline and inspect confounding.

03

## Days 61-90

Prepare pilot evidence and failure analysis.

Set sample-size and external-validation plans with a statistician.

Decide whether to expand, revise the endpoint or stop.

**Decision requested from collaborators: a clinical lead, a data/assay partner and support for the pilot study.**

# Reproducible starting point for collaborators

Implementation is available in the pancreatic-image-analysis project folder.

| Purpose | Files | What collaborators can review |
|---|---|---|
| Run the prototype | README.md; run_demo.ps1; requirements.txt | Local setup and synthetic workflow; model artifacts remain research-only. |
| Inspect the analysis | pipeline.py; make_demo.py; tests/test_pipeline.py | Feature extraction, training, inference and meaningful software checks. |
| Review the product | app.py; context.py; outputs/*.json | Local dashboard, independent treatment context and traceable demo reports. |
| Plan the study | docs/PROTOCOL.md; docs/MODEL_CARD.md | Research questions, limitations, validation gates and development scope. |

# Sources, evidence status and interpretation

Sources checked 13 November 2026. Project proposals are planning recommendations.

**[1] NCI. Pancreatic Cancer Treatment (PDQ), professional summary.**

https://www.cancer.gov/types/pancreatic/hp/pancreatic-treatment-pdq

**[2] Yang et al. Guidelines and definitions for research on EMT. Nature Reviews Molecular Cell Biology, 2020.**

https://www.nature.com/articles/s41580-020-0237-9

**[3] FDA. AI-enabled device software functions: lifecycle management and marketing submission recommendations. January 2025. DRAFT.**

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/artificial-intelligence-enabled-device-software-functions-lifecycle-management-and-marketing

## Local evidence

Demo metrics and example values come from outputs/metrics.json and outputs/prediction.json. The example prediction uses an in-training artificial sample. Four automated tests and a dashboard smoke test passed. Clinical performance, utility and regulatory readiness remain unestablished.